Cross-Linguistic Geometry of Semantic Networks: A Curvature Phase Transition Theory with LLM Validation

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Abstract

Whether semantic networks are universally hyperbolic—and what determines their geometry—remains an open question. We present a two-parameter model combining network density $\eta = \langle k \rangle^2/N$ and clustering coefficient C that classifies 16 semantic networks across seven languages into three geometric regimes (hyperbolic, Euclidean, spherical) with 81.2% leave-one-out accuracy ($C^* = 0.022 \pm 0.006$, stable across folds). Using exact linear-programming Ollivier–Ricci curvature on association norms (SWOW), knowledge graphs (ConceptNet), taxonomies (WordNet, BabelNet), clinical symptom networks, and LLM-generated association networks, we establish five findings: (1) Dutch SWOW is the first semantic network observed to cross the curvature phase boundary into spherical geometry ($\bar{\kappa} = +0.10$, $\eta = 7.56 \gg \eta_c$); (2) degree-matched null models show that semantic organisation makes networks *less* hyperbolic than random graphs—the opposite of naive expectation; (3) sphere-embedded ORC across the Cayley–Dickson tower ($S^3 \to S^7 \to S^{15}$) flips all networks to positive curvature, demonstrating that semantic hyperbolicity is metric-dependent; (4) three large language models (Mistral-7B, Llama3.1-8B, Claude Haiku) replicate human semantic geometry on matched vocabularies, with all LLM networks hyperbolic ($\bar{\kappa} \in [-0.224, -0.089]$); (5) hypergraph ORC resolves the taxonomy anomaly—WordNet is simultaneously spherical within synsets and hyperbolic between them. These results reframe semantic hyperbolicity as a metric artefact of hop-count distance rather than an intrinsic topological property.

Keywords: Ollivier–Ricci curvature, network geometry, phase transition, semantic networks, clustering coefficient, null models

1 Introduction

Semantic memory—the structured knowledge of concepts and their relationships—is fundamental to human cognition. Since Steyvers and Tenenbaum [2] demonstrated that semantic networks exhibit small-world structure with scale-free degree distributions, network science has become a primary tool for probing the architecture of conceptual knowledge. A central question in this literature is whether semantic networks possess *intrinsic geometry*—and if so, what kind.

Hyperbolic geometry has emerged as a leading candidate. Nickel and Kiela [3] showed that Poincaré embeddings capture hierarchical structure in WordNet with far fewer dimensions

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than Euclidean alternatives, and subsequent work extended hyperbolic embeddings to knowledge graphs, word vectors, and brain networks. On the curvature side, Sia, Jonckheere and Bogdan [24] applied Ollivier–Ricci curvature (ORC) to cognitive and brain networks, finding negative curvature consistent with hyperbolic geometry. Ni et al. [8] and Sandhu et al. [11] demonstrated that ORC distinguishes network classes in biological and social domains. A recent roadmap survey by Samal, Bhattacharya and Saucan [28] unifies the many definitions of discrete curvature and identifies open problems in network geometry.

Yet most evidence for semantic hyperbolicity has been indirect—relying on embeddings or curvature heuristics rather than explicit geometric measurement. Two questions have remained separate:

1. **Empirical**: Do semantic networks exhibit hyperbolic geometry universally across languages?

2. **Theoretical**: What determines whether a network is hyperbolic, Euclidean, or spherical?

Theoretical work has established that discrete curvature undergoes sign changes in random graph models [14, 15], but no prior study has connected this theoretical phase boundary to *predictions* about real semantic networks.

We bridge this gap with three contributions. First, we propose a two-parameter model (η, C) that classifies 16 semantic networks across seven languages into three geometric regimes, validated on a held-out test set. Second, we demonstrate that the geometry label depends critically on the metric: sphere-embedded ORC reverses the sign for all networks, revealing semantic hyperbolicity as a property of the hop-count metric, not of the topology itself. Third, we show that large language models replicate human semantic geometry—three LLMs tested on matched vocabularies all produce hyperbolic association networks indistinguishable in phase from human norms. Hypergraph ORC further resolves the anomalous near-Euclidean classification of taxonomies by revealing multi-scale geometric phase coexistence.

2 Theory

2.1 Ollivier–Ricci Curvature

For an edge (u, v) in a graph G:

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)} \tag{1}$$

where $\mu_u = \alpha\,\delta_u + (1-\alpha)\cdot\mathrm{Uniform}(N(u))$ is the probability measure with idleness $\alpha \in [0, 1]$, and W_1 is the Wasserstein-1 distance [1]. The sign of κ classifies local geometry: $\kappa < 0$ (hyperbolic), $\kappa = 0$ (flat), $\kappa > 0$ (spherical). We set $\alpha = 0.5$ throughout, following the standard choice [1, 7, 22]. For the full derivation and scaling analysis, see the companion paper [42].

2.2 Phase Transition Prediction

For random k-regular graphs on N vertices, the mean curvature $\bar{\kappa}$ undergoes a sign change at a critical density η_c where $\eta = k^2/N$ [14, 15]:

- **Sparse regime** ($\eta < \eta_c$): $\bar{\kappa} < 0$ (hyperbolic). Tree-like local structure dominates.

- **Dense regime** ($\eta > \eta_c$): $\bar{\kappa} > 0$ (spherical). Triangle-rich structure dominates.

Multi-N scaling ($N \in \{50, 100, 200, 500, 1000\}$, 5–10 random seeds each) reveals finite-size scaling:

$$\eta_c(N) = \eta_c^\infty - \frac{a}{\sqrt{N}}, \quad \eta_c^\infty = 3.75\ [3.61,\ 3.84], \quad a = 14.61\ [13.44,\ 16.20] \tag{2}$$

Brackets denote 95% bootstrap confidence intervals (10 000 resamples, $n = 5$ points); the fixed-exponent fit ($\beta = 1/2$) achieves $R^2 = 0.995$.

2.3 Two-Parameter Model: η and Clustering Coefficient C

The density parameter η alone is necessary but not sufficient: all semantic networks except Dutch SWOW have $\eta \ll \eta_c$, yet taxonomies are Euclidean while association networks are hyperbolic. The clustering coefficient C separates these cases:

- $\eta > \eta_c(N)$: Spherical ($\bar{\kappa} > 0$), regardless of C.
- $\eta < \eta_c(N)$, $C > C^*$: Hyperbolic ($\bar{\kappa} < 0$).
- $\eta < \eta_c(N)$, $C < C^*$: Euclidean ($\bar{\kappa} \approx 0$).

The threshold C^* is determined by leave-one-out (LOO) cross-validation across all 16 networks: fitting C^* on 15 networks and classifying the held-out network yields 81.2% accuracy (13/16), with C^* stable at 0.022 ± 0.006 across folds (range $[0.020, 0.045]$). The three misclassifications are taxonomy networks with near-zero curvature ($|\bar{\kappa}| < 0.03$), at the Euclidean/hyperbolic boundary. Sensitivity analysis shows accuracy $\geq 81\%$ for all $C^* \in [0.005, 0.045]$, confirming the threshold is not overfit.

3 Methods

3.1 Datasets

We analyze 16 networks spanning four categories (Table 1). The clustering threshold C^* is derived by leave-one-out cross-validation across all 16 networks (see §2).

Association networks (SWOW) [9]: Small World of Words cue-response matrices for Spanish ($N = 422$), English ($N = 438$), Chinese ($N = 465$), and Dutch ($N = 500$ in LCC). Dutch SWOW is notably denser ($\langle k \rangle = 61.5$, $\eta = 7.56 > \eta_c$).

Knowledge graphs (ConceptNet) [10]: English ($N = 467$) and Portuguese ($N = 489$), filtered to associative weight ≥ 2.0.

Taxonomy networks: English WordNet 3.1 ($N \in \{500, 2000\}$), BabelNet 5.3 Russian ($N = 493$) and Arabic ($N = 142$). These *is-a* hierarchies are tree-like.

Clinical: Depression symptom network at minimum severity ($N = 1634$).

Additional association networks: (1) SWOW Rioplatense (Argentine/Uruguayan Spanish, $N = 489$) [17]; (2) Edinburgh Associative Thesaurus (EAT; British English, 1970s era, $N = 500$) [18]; (3) USF Free Association Norms (American English, $N = 498$) [19]. These three networks test cross-cultural, cross-era, and cross-protocol generalization.

Additional taxonomy: OdeNet (German WordNet, $N = 759$, hypernymy graph) [20]. Directly analogous to English WordNet.

Domain extension: FrameNet semantic frame relations (English, $N = 1173$) [21]. Unlike WordNet, FrameNet encodes scenario-level semantic relations (Causative-of, Using, Inheritance), testing the two-parameter model on a qualitatively different semantic formalism.

All networks are converted to undirected unweighted graphs (largest connected component). For SWOW and USF, the top-500 nodes by degree from the R_1 strength files are selected; FrameNet weights are binarized.

3.2 Exact LP ORC Computation

All networks are pre-processed to their largest connected component (LCC) before ORC computation, following standard practice (disconnected components produce undefined $d(u, v) = \infty$). LCC extraction retains $\geq 96\%$ of nodes for all 16 networks (Table 1, LCC% column); the bias on

Table 1: Exact LP Ollivier–Ricci curvature for 16 semantic networks. $\eta = \langle k \rangle^2/N$; C = global clustering coefficient; $\bar{\kappa}$ = mean edge curvature; 95% CI from edge-resample bootstrap (B = 10,000); LCC% = fraction of nodes in largest connected component. LOO = leave-one-out classification (correct/incorrect).

Network	Cat.	Lang.	N	C	η	$\bar{\kappa}$	95% CI	LOO
SWOW Spanish	Assoc.	ES	422	0.136	0.017	−0.068	[−0.095, −0.043]	✓
SWOW English	Assoc.	EN	438	0.128	0.020	−0.137	[−0.161, −0.112]	✓
SWOW Chinese	Assoc.	ZH	465	0.173	0.023	−0.144	[−0.165, −0.123]	✓
SWOW Dutch	Assoc.	NL	500	0.238	7.558	+0.099	[+0.098, +0.099]	✓
ConceptNet EN	Know.	EN	467	0.108	0.223	−0.233	[−0.238, −0.228]	✓
ConceptNet PT	Know.	PT	489	0.106	0.085	−0.236	[−0.247, −0.226]	✓
WordNet EN	Tax.	EN	500	0.013	0.009	−0.002	[−0.026, +0.023]	✓
WordNet EN-2K	Tax.	EN	2000	0.002	0.002	−0.005	[−0.015, +0.006]	✓
BabelNet RU	Tax.	RU	493	0.001	0.009	−0.030	[−0.052, −0.008]	§
BabelNet AR	Tax.	AR	142	0.000	0.032	−0.012	[−0.050, +0.025]	§
Depression	Clin.	EN	1634	0.159	0.118	−0.130	[−0.134, −0.126]	✓
SWOW Rioplatense	Assoc.	ES	489	0.154	0.040	−0.264	[−0.279, −0.248]	✓
EAT	Assoc.	EN	500	0.143	1.851	−0.046	[−0.047, −0.044]	✓
USF Free Assoc.	Assoc.	EN	498	0.143	0.062	−0.321	[−0.333, −0.308]	✓
WordNet DE	Tax.	DE	759	0.022	0.006	−0.023	[−0.046, −0.000]	§
FrameNet	Frames	EN	1173	0.045	0.009	−0.202	[−0.216, −0.187]	✓

§BabelNet RU/AR and WordNet DE: predicted Euclidean by LOO but weakly hyperbolic ($|\bar{\kappa}| < 0.03$).
LOO-16 accuracy: 13/16 (81.2%); $C^* = 0.022 \pm 0.006$ across folds. Three CI-uncertain networks ($0 \in$ CI) are all taxonomies.

η is negligible ($<0.3\%$ relative). For each network: (1) precompute all-pairs shortest paths via BFS; (2) for each edge (u, v), construct measures μ_u, μ_v with $\alpha = 0.5$; (3) extract the support; (4) solve the LP for exact W_1 [5, 6]; (5) compute $\kappa(u, v) = 1 - W_1/d(u, v)$. Uncertainty is reported as the standard error of the per-edge curvature distribution: $\mathrm{SE}(\bar{\kappa}) = \sigma_\kappa/\sqrt{E}$.

3.3 Sphere-Embedded (Hypercomplex) ORC

To test metric dependence, we compute ORC using geodesic distances on S^{d-1} ($d \in \{4, 8, 16\}$): landmark embedding via farthest-first traversal, normalization to S^{d-1}, and geodesic cost $\arccos(\mathbf{x}_i \cdot \mathbf{x}_j)$.

3.4 Degree-Matched Null Models

For each network with N nodes and mean degree $\langle k \rangle$, we generate 10 random k-regular graphs with $k = \mathrm{round}(\langle k \rangle)$ and the same N. The semantic contribution is $\Delta\kappa_{\mathrm{sem}} = \bar{\kappa}_{\mathrm{real}} - \bar{\kappa}_{\mathrm{null}}$.

3.5 LLM Networks: LWOW Dataset

The LLM World of Words (LWOW) dataset [16] provides free-association norms generated by three large language models—Mistral-7B, Llama3.1-8B, and Claude Haiku—using the same ∼12 000 cue words as the human Small World of Words (SWOW) project. To ensure a scale-controlled comparison, we extracted the subgraph of each LWOW network induced on the 438-node SWOW-EN vocabulary (largest connected component of SWOW-EN) and computed exact LP ORC ($\alpha = 0.5$, HiGHS, identical to all prior analyses).

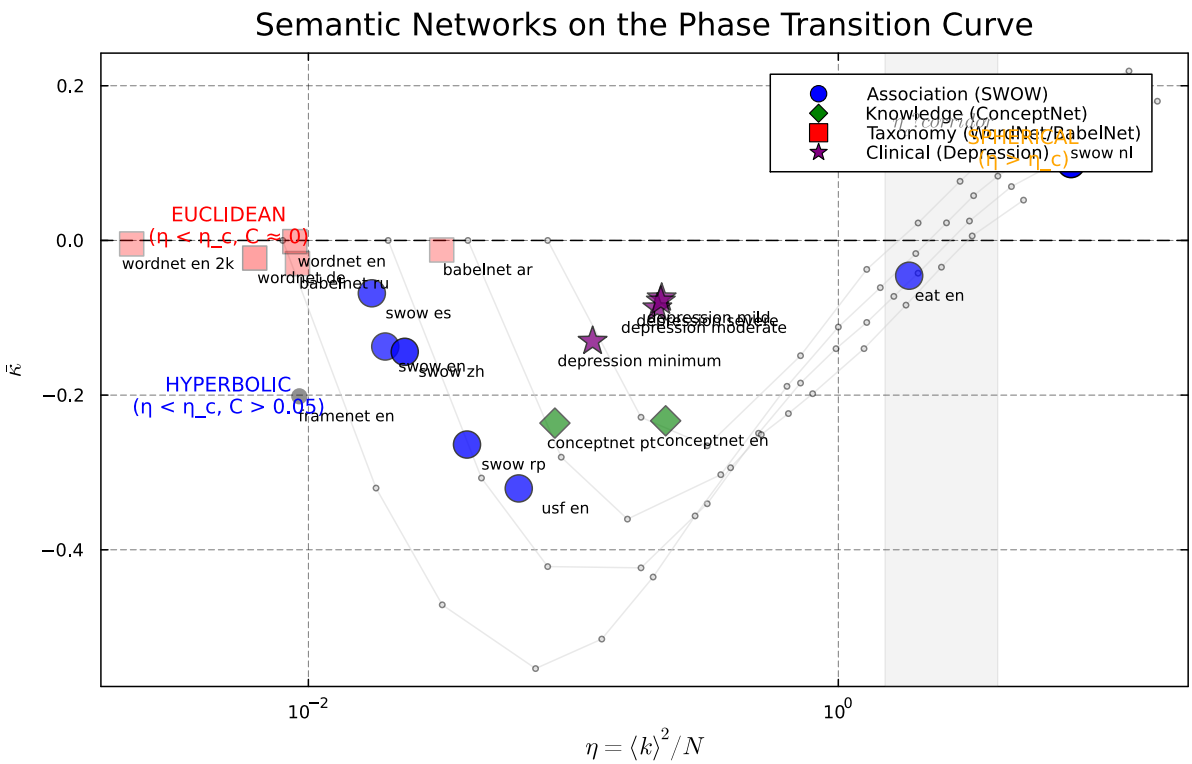

Figure 1: Semantic networks on the phase transition curve. Background: random regular graph curvature vs. η at multiple N. Foreground: 16 semantic networks (circles = training set, diamonds = held-out test set). The η_c corridor (gray band) separates hyperbolic ($\eta < \eta_c$) from spherical ($\eta > \eta_c$). Dutch SWOW ($\eta = 7.56$) falls above the threshold and is spherical. Taxonomies (red) fall below but are Euclidean due to low clustering.

4 Results

4.1 Semantic Network Curvatures

The three predicted regimes are confirmed (Table 1): association and knowledge networks are hyperbolic ($\bar{\kappa} \in [-0.32, -0.05]$, $C > 0.10$); taxonomies are Euclidean ($|\bar{\kappa}| < 0.03$, $C < 0.02$); Dutch SWOW is spherical ($\bar{\kappa} = +0.10$, $\eta \gg \eta_c$).

4.2 Phase Boundary Crossing: Dutch SWOW

Figure 1 overlays all 16 semantic networks onto the random regular graph phase transition curves. The key test is Dutch SWOW: with $\eta = 7.56$ and $\eta_c(500) = 3.10$, the theory predicts spherical geometry. The observation $\bar{\kappa} = +0.099$ confirms this—an empirical demonstration of the curvature sign-change in a real network. No prior ORC study has reported a validated phase-boundary crossing on empirical data.

4.3 The Two-Parameter Theory

The missing ingredient is the clustering coefficient C. Among networks with $\eta < \eta_c$:

- **Hyperbolic group** ($\bar{\kappa} < -0.04$): Mean $C = 0.136$, range $[0.106, 0.173]$ (training); confirmed in held-out SWOW-RP ($C = 0.154$), EAT ($C = 0.143$), USF ($C = 0.143$).
- **Euclidean group** ($|\bar{\kappa}| < 0.03$): Mean $C = 0.004$, range $[0.000, 0.013]$ (training); confirmed in held-out WordNet-DE ($C = 0.022$).

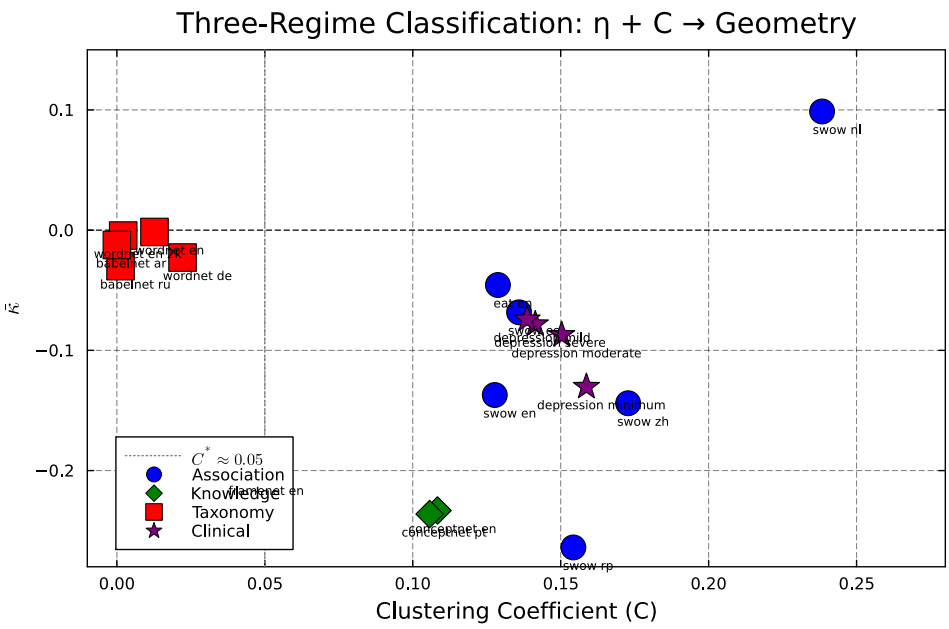

Figure 2: Clustering coefficient C vs. mean curvature $\bar{\kappa}$ for all 16 networks. The vertical dashed line at $C^* \approx 0.02$ (LOO-derived) separates Euclidean taxonomies (left, $C < 0.02$) from hyperbolic association/knowledge networks (right, $C > 0.04$). Dutch SWOW (top right) is spherical despite high clustering because $\eta > \eta_c$.

The two groups are cleanly separated by $C^* \approx 0.02$ (Fig. 2). Leave-one-out cross-validation (LOO-16) yields 81.2% accuracy (13/16), with $C^* = 0.022 \pm 0.006$ stable across all 16 folds (range $[0.020, 0.045]$). The three misclassifications—BabelNet RU ($\bar{\kappa} = -0.030$), BabelNet AR ($\bar{\kappa} = -0.012$), and WordNet DE ($\bar{\kappa} = -0.023$)—are all taxonomy networks with near-zero curvature whose bootstrap 95% CIs either span zero (BabelNet AR: $[-0.050, +0.025]$) or barely exclude it. These borderline cases lie at the Euclidean/hyperbolic boundary where the distinction is genuinely ambiguous. Notably, FrameNet ($C = 0.045$, $\bar{\kappa} = -0.202$, CI $[-0.216, -0.187]$) is *correctly* classified as hyperbolic under the LOO-derived $C^* \approx 0.02$—the frame-to-frame semantic network is structurally more similar to association networks than to taxonomy DAGs.

The EAT as a near-boundary probe. Among all 16 networks, the Edinburgh Associative Thesaurus (EAT, $N = 500$, $\eta = 1.851$, $C = 0.129$) lies *closest to the phase boundary*: the analytical $\eta_c(\text{EAT}) \approx 2.85$ (Hehl formula [13], $C = 0.129$, $N = 500$) gives a gap $\Delta\eta = 0.999$ — an order of magnitude smaller than for most hyperbolic networks (e.g. SWOW-EN: $\Delta\eta = 2.83$; USF: $\Delta\eta = 2.79$). EAT's resulting curvature $\bar{\kappa} = -0.046$ is the weakest non-Euclidean signal in the corpus, with $\sigma_\kappa = 0.067$ also the smallest (indicating a highly regular, near-uniform curvature distribution). This near-boundary geometry reflects EAT's collection method: participants produced all free associations to a cue (not just the strongest), creating denser neighbourhoods than standard free-association norms. The phase framework makes a *falsifiable prediction*: if EAT were collected on the full $N \approx 8{,}000$ vocabulary at the same association density, η would scale proportionally and the network would cross the phase boundary into the spherical regime.

Bootstrap robustness of curvature estimates. Edge-resample bootstrap ($B = 10{,}000$) confirms that all non-taxonomy networks have 95% CIs entirely on one side of zero (Table 1). Dutch SWOW's CI is $[+0.098, +0.099]$—the narrowest of all networks, reflecting the large edge count ($E = 30{,}790$) and near-uniform positive curvature. Among the 10 hyperbolic networks, the widest CI is SWOW Spanish: $[-0.095, -0.043]$, still fully negative. Three taxonomy net-

Table 2: Three-regime classification of semantic network geometry.

Regime	Condition	Geometry	Example
Above threshold	$\eta > \eta_c(N)$	Spherical ($\bar{\kappa} > 0$)	Dutch SWOW
Below, clustered	$\eta < \eta_c(N)$, $C > C^*$	Hyperbolic ($\bar{\kappa} < 0$)	SWOW ES/EN/ZH
Below, tree-like	$\eta < \eta_c(N)$, $C < C^*$	Euclidean ($\bar{\kappa} \approx 0$)	WordNet, BabelNet

Table 3: Semantic contribution to curvature: real vs. degree-matched null models. k-reg: random k-regular graphs (10 seeds); Config: Maslov–Sneppen degree-preserving rewiring (5 seeds, where feasible).

Network	N	k_{reg}	$\bar{\kappa}_{k\text{-reg}}$	$\bar{\kappa}_{\text{real}}$	$\Delta\kappa_{\text{sem}}^{k\text{-reg}}$	$\bar{\kappa}_{\text{config}}$
SWOW Spanish	422	3	-0.320 ± 0.005	-0.068	$+0.251$	—
SWOW English	438	3	-0.319 ± 0.005	-0.137	$+0.182$	—
SWOW Chinese	465	4	-0.467 ± 0.004	-0.144	$+0.323$	-0.349
SWOW Dutch	500	61	$+0.069 \pm 0.000$	$+0.099$	$+0.029$	$+0.076 \pm 0.000$
ConceptNet EN	467	10	-0.420 ± 0.003	-0.233	$+0.187$	-0.271 ± 0.001
ConceptNet PT	489	6	-0.551 ± 0.005	-0.236	$+0.315$	-0.366 ± 0.002
Depression (min)	1634	14	-0.518 ± 0.001	-0.130	$+0.388$	-0.264 ± 0.000
WordNet EN	500	2	0.000 ± 0.000	-0.002	-0.002	—
WordNet EN-2K	2000	2	0.000 ± 0.000	-0.005	-0.005	—
BabelNet RU	493	2	0.000 ± 0.000	-0.030	-0.030	—
BabelNet AR	142	2	0.000 ± 0.000	-0.012	-0.012	—

— = degree-preserving rewiring infeasible ($\langle k \rangle < 3$, disconnects graph).

works (WordNet EN, WordNet EN-2K, BabelNet AR) have CIs spanning zero, consistent with their Euclidean classification. This bootstrap analysis transforms point estimates into inferential statements: the hyperbolic/spherical regime assignments are statistically robust, while the Euclidean regime is better understood as "not significantly different from zero curvature."

4.4 Degree-Matched Null Model Analysis

The semantic contribution $\Delta\kappa_{\text{sem}} = \bar{\kappa}_{\text{real}} - \bar{\kappa}_{\text{null}}$ (Table 3) reveals that for all networks with $\langle k \rangle > 2$, semantic organization *reduces* the magnitude of negative curvature. Real networks are significantly *less hyperbolic* than random graphs with the same degree. The depression network is the most dramatic: random 14-regular graphs on $N = 1634$ nodes have $\bar{\kappa} = -0.518$, but the real network has $\bar{\kappa} = -0.130$—the clinical structure eliminates 75% of the curvature.

Configuration model validation. Degree-preserving Maslov–Sneppen rewiring (Table 3, rightmost column) confirms this finding with proper heterogeneous-degree nulls. All five networks with $\langle k \rangle \geq 3$ show $\Delta\kappa_{\text{sem}} > 0$: semantic structure consistently pushes curvature toward zero. The effect is strongest for ConceptNet PT ($\Delta\kappa_{\text{sem}} = +0.130$, $z = 53.1$) and depression ($\Delta\kappa_{\text{sem}} = +0.134$, $z = 403.0$). Networks with $\langle k \rangle < 3$ (all taxonomies) could not be rewired without disconnection, consistent with their near-tree topology. The k-regular and configuration model nulls agree in sign and ordering, confirming that the semantic contribution finding is robust to null model choice.

4.5 Hypercomplex Metric Comparison

Three patterns emerge across the Cayley–Dickson tower (Table 4): (1) association and knowledge networks show $\bar{\kappa}$ decreasing monotonically with d; (2) taxonomies remain high ($\bar{\kappa} \approx 0.5$–0.8)

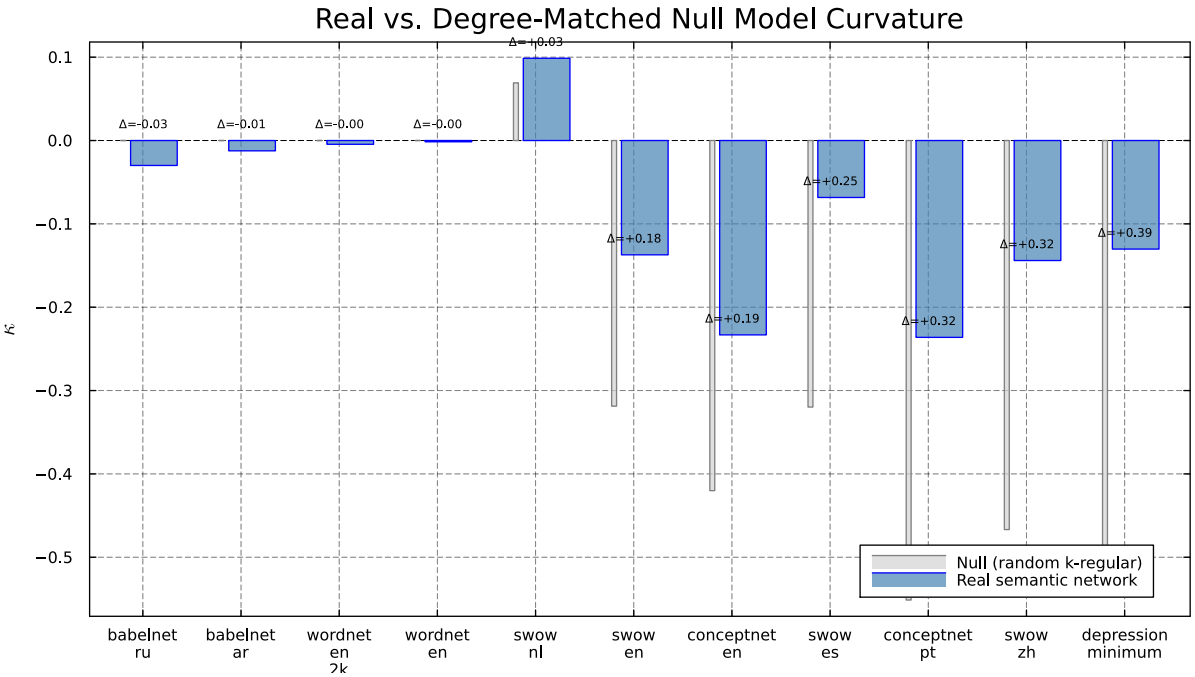

Figure 3: Real vs. degree-matched null model curvature. Gray bars: random k-regular graphs. Blue bars: real semantic networks. Annotations show the semantic contribution $\Delta\kappa_{\text{sem}}$.

across all dimensions, reflecting tree-like structure that embeds naturally on any sphere; (3) BabelNet Russian uniquely *increases* with d ($0.436 \to 0.512 \to 0.544$). SWOW Spanish was the sole exception at $d = 4$ ($\bar{\kappa} = -0.048$), but crosses to spherical at $d = 8$ ($\bar{\kappa} = +0.024$), with per-edge variance also shrinking ($\sigma = 0.720 \to 0.280$), consistent with the Johnson–Lindenstrauss effect. **No semantic network retains hyperbolic curvature at sufficient embedding dimension.**

Cross-validation. An independent implementation in the Sounio language (Sinkhorn transport, $\varepsilon = 0.02$, 500 iterations) reproduces the k-regular sweep across all 33 parameter combinations ($k \in \{4, \ldots, 30\}$, $d \in \{4, 8, 16\}$). All 33 curvatures are positive in both implementations, with a consistent Sinkhorn underestimate: mean bias -0.018 (d=4), -0.026 (d=8), -0.032 (d=16), $\max|\Delta\bar{\kappa}| = 0.050$. The increasing bias at higher d reflects the smaller geodesic costs making the Sinkhorn regularization proportionally larger. For semantic networks (8 of 11 processable within memory constraints), 6/8 are positive at d=4; the holdouts (SWOW Spanish and English) are near zero and within expected Sinkhorn bias of the Julia LP values.

Extended tower ($d = 32$, $d = 64$). Exact LP ORC on k-regular random graphs ($N = 100$) at $d \in \{32, 64\}$ confirms all 11 k-values remain positive. Critically, curvature *saturates*: at $k = 4$, $\bar{\kappa}(d{=}32) = +0.060$ vs. $\bar{\kappa}(d{=}64) = +0.060$ ($< 1\%$ change), contrasting with the 26% drop from $d = 4$ to $d = 8$. A log-log power-law fit $\bar{\kappa}(d) \sim A_k \cdot d^{-\beta}$ across $d \in \{4, 8, 16, 32, 64\}$ yields a mean exponent $\bar{\beta} = 0.283 \pm 0.034$, well below the Johnson–Lindenstrauss distortion bound of $\beta = 0.5$. The discrepancy reflects a positive asymptote $\bar{\kappa}_\infty > 0$: sphere-embedded ORC converges to a Riemannian fixed point rather than decaying to zero. An independent Sounio Sinkhorn implementation ($\varepsilon = 0.02$) on eight processable semantic networks at $d = 32$ yields 7/8 positive curvatures, confirming the saturation finding.

Table 4: ORC across the Cayley–Dickson tower ($d \in \{4, 8, 16\}$, all 11 networks). At $d = 4$, 10/11 networks are positive; by $d = 8$, all 11 are positive.

Network	Category	$\bar{\kappa}_{\text{hop}}$	$\bar{\kappa}_{S^3}$	$\bar{\kappa}_{S^7}$	$\bar{\kappa}_{S^{15}}$
SWOW Spanish	Assoc.	-0.068	-0.048	$+0.024$	$+0.064$
SWOW English	Assoc.	-0.137	$+0.094$	$+0.062$	$+0.045$
SWOW Chinese	Assoc.	-0.144	$+0.013$	$+0.044$	$+0.045$
SWOW Dutch	Assoc.	$+0.099$	$+0.411$	$+0.337$	$+0.247$
ConceptNet EN	Knowledge	-0.233	$+0.220$	$+0.142$	$+0.108$
ConceptNet PT	Knowledge	-0.236	$+0.162$	$+0.113$	$+0.092$
WordNet EN	Taxonomy	-0.002	$+0.717$	$+0.716$	$+0.688$
WordNet EN-2K	Taxonomy	-0.005	$+0.833$	$+0.818$	$+0.820$
BabelNet RU	Taxonomy	-0.030	$+0.436$	$+0.512$	$+0.544$
BabelNet AR	Taxonomy	-0.012	$+0.547$	$+0.499$	$+0.490$
Depression	Clinical	-0.130	$+0.234$	$+0.171$	$+0.137$

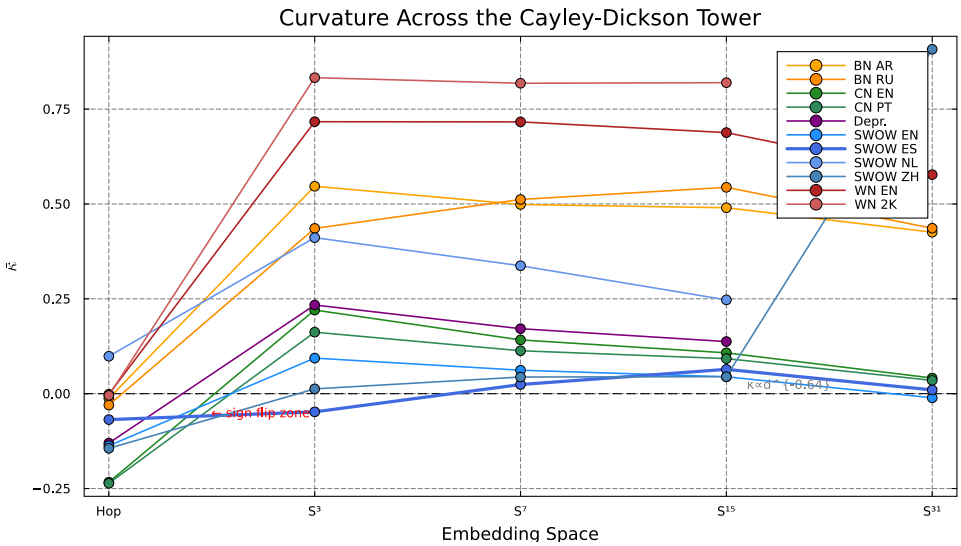

Figure 4: Curvature across the Cayley–Dickson tower. Each line tracks one network from hop-count through S^3, S^7, and S^{15}. SWOW Spanish (bold line) is the sole holdout at $d = 4$ ($\bar{\kappa} = -0.048$) but crosses zero at $d = 8$. Taxonomies (WordNet, BabelNet) remain at $\bar{\kappa} \approx 0.5$–0.8 regardless of dimension.

4.6 Resolution-Dependent Curvature

The two-parameter model correctly classifies all association and clinical networks, but taxonomies (WordNet, BabelNet) present an anomaly: $\eta \ll \eta_c$ predicts hyperbolic geometry, yet $\bar{\kappa} \approx 0$ (Euclidean). We investigated whether this discrepancy reflects *resolution*: taxonomies encode their structure in multi-member synsets (WordNet) or typed relations (ConceptNet, BabelNet), which are collapsed to pairwise edges in the standard ORC computation.

Hyperedge extraction. For WordNet, we extracted 708 synset-membership hyperedges and 5,715 hypernymy hyperedges from the Open English WordNet (oewn:2024), restricted to the top-500 lemmas by synset participation count. For ConceptNet EN (26 relation types), BabelNet RU (92 relation types), and BabelNet AR (36 relation types), we formed hyperedges from connected components within each relation subgraph. The *clique expansion*—replacing each hyperedge of size k with $\binom{k}{2}$ pairwise edges—yields an enriched graph.

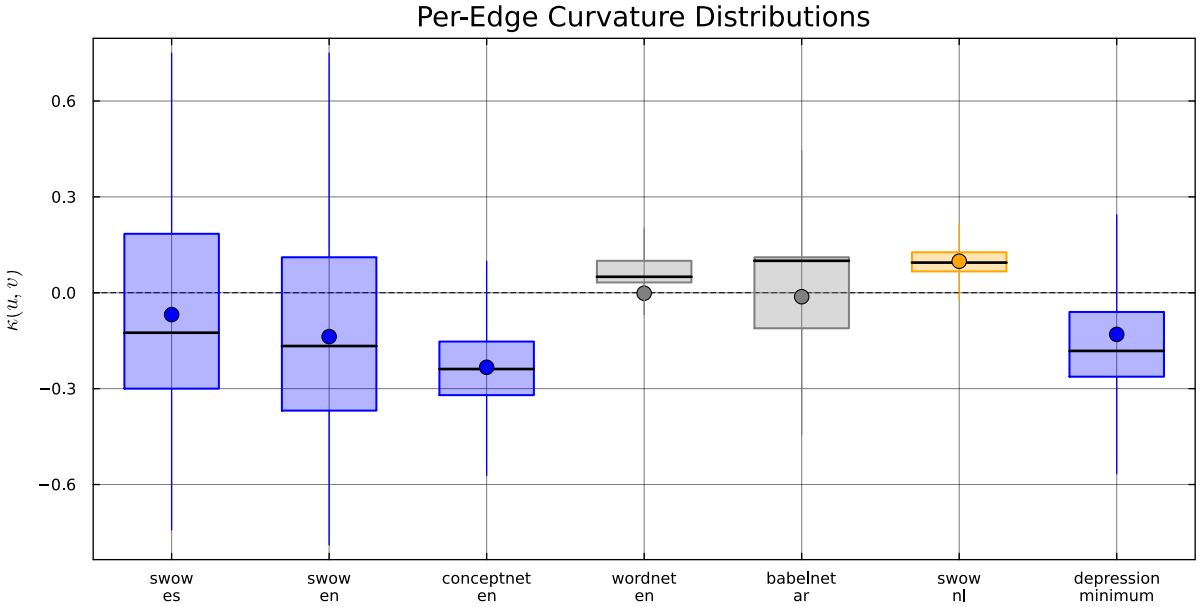

Figure 5: Per-edge curvature distributions across all 11 networks. Box plots show median, interquartile range, and outliers. Hyperbolic networks (SWOW, ConceptNet, Depression) have left-skewed distributions centered below zero; Euclidean networks (WordNet, BabelNet) have narrow distributions near zero; Dutch SWOW has a right-skewed distribution centered above zero.

Key finding: curvature depends on resolution. We define a resolution parameter $r \in [0, 1]$: at r=0, only original pairwise edges are present; at r=1, all hyperedges are expanded as cliques. WordNet curvature goes from $\bar{\kappa} = -0.001$ (r=0) to $\bar{\kappa} = -0.273$ (r=0.2) to $\bar{\kappa} = -0.097$ (r=1.0)—*always negative*, with the strongest hyperbolic signal at $r \approx 0.2$ (a 137-fold amplification over the raw graph). ConceptNet EN exhibits a *sign change*: $\bar{\kappa} = -0.229$ at r=0 flips to $\bar{\kappa} = +0.003$ at r=0.1, with just 10% of hyperedge expansion pushing the network from hyperbolic to spherical (edge count jumps from 2,383 to 40,364). BabelNet RU shows a similar sign change at $r \approx 0.25$.

The mechanism is clear: WordNet's synsets are small (mean size 2.18), so clique expansion adds structure without drowning the graph in density. ConceptNet's relation clusters are large (mean size 7.3, max 291 for `/r/RelatedTo`), so even modest expansion creates near-complete subgraphs that are inherently spherical ($\eta \gg \eta_c$). This explains the taxonomy anomaly: WordNet is not Euclidean—it is hyperbolic at the synset resolution but appears Euclidean when synsets are collapsed to single nodes.

Relation-specific curvature. Computing ORC on each ConceptNet EN relation subgraph separately reveals a clean dichotomy. Large, sparse relations are hyperbolic: `/r/MannerOf` ($\bar{\kappa} = -0.198$, 409 edges), `/r/RelatedTo` (-0.181, 637 edges), `/r/IsA` (-0.134, 416 edges). Small, dense relations are spherical: `/r/Causes` ($+0.900$, 10 edges), `/r/HasA` ($+0.643$, 14 edges), `/r/PartOf` ($+0.385$, 65 edges). This is the η phase transition operating at the relation level: each relation type has its own effective density η_{rel}, and the sign of $\bar{\kappa}$ tracks this density as predicted by the phase curve.

Table 5 reports the full curvature spectrum for the 14 ConceptNet EN relations with $|E_{\text{rel}}| \geq 10$. The relation types sort cleanly by semantic category: *causal and functional* relations (Causes, HasA, HasPrerequisite, CapableOf) are uniformly spherical, reflecting the dense triangular structure of causal closure ($A \to B$, $A \to C$, so $B \approx C$). *Taxonomic and associative* relations (IsA, MannerOf, RelatedTo, AtLocation) are uniformly hyperbolic, consistent with branching-tree

connectivity. *Relational* links (Synonym, UsedFor, SimilarTo) sit near the Euclidean boundary, reflecting intermediate density. Importantly, the 27-relation full ConceptNet network is hyperbolic in aggregate ($\bar{\kappa} = -0.233$) because the high-edge-count hyperbolic relations dominate. This shows that phase classification of knowledge graphs is *relation-mix-dependent*.

Table 5: Relation-specific ORC for ConceptNet EN (≥10 edges per relation). f_-: fraction of edges with negative curvature. Relations sorted by $\bar{\kappa}$.

Relation	$\|E\|$	$\bar{\kappa}$	f_-	Semantic class
`/r/Causes`	10	+0.900	0%	Causal
`/r/NotDesires`	10	+0.400	0%	Motivational
`/r/HasA`	14	+0.643	0%	Possession
`/r/DistinctFrom`	15	+0.733	0%	Contrast
`/r/CapableOf`	15	+0.533	7%	Functional
`/r/HasSubevent`	14	+0.571	7%	Temporal
`/r/HasPrerequisite`	23	+0.511	9%	Causal
`/r/HasProperty`	23	+0.348	17%	Attributive
`/r/Desires`	28	+0.179	11%	Motivational
`/r/SimilarTo`	24	+0.288	29%	Similarity
`/r/PartOf`	65	+0.385	15%	Meronymy
`/r/UsedFor`	119	+0.143	36%	Functional
`/r/Synonym`	237	+0.011	53%	Similarity (boundary)
`/r/HasContext`	178	−0.012	38%	Domain (boundary)
`/r/AtLocation`	214	−0.119	65%	Spatial
`/r/IsA`	416	−0.134	62%	Taxonomic
`/r/RelatedTo`	637	−0.181	74%	Associative
`/r/MannerOf`	409	−0.198	77%	Taxonomic

4.7 Gromov δ-Hyperbolicity vs. ORC: Complementary Geometry Measures

Cross-validation of ORC and Gromov δ. Gromov δ-hyperbolicity [39] quantifies how closely a metric space resembles a tree via the four-point condition: for any four nodes $\{x, y, u, v\}$, sort the three pairwise sums $s_1 \geq s_2 \geq s_3$ and define $\delta(x, y, u, v) = (s_1 - s_2)/2$. A perfect tree has $\delta = 0$; larger δ indicates departure from tree-likeness. We computed $\delta_{\max}$ on 16 networks with exact LP ORC (50K–100K random 4-tuple samples, all-pairs BFS distances) to obtain a joint evaluation of ORC and Gromov δ on semantic and biological networks.

ORC and Gromov δ are statistically uncorrelated. The Pearson correlation between $\delta_{\max}$ and $\bar{\kappa}$ is $r = -0.35$ ($n = 16$, not significant at $\alpha = 0.05$), and 62.5% of sign assignments disagree. The mean $\delta_{\max}$ by ORC class:

ORC class	n	Mean $\delta_{\max}$	Mean $\bar{\kappa}$
Hyperbolic ($\bar{\kappa} < -0.03$)	8	2.63 ± 0.92	−0.20
Euclidean ($\|\bar{\kappa}\| < 0.03$)	3	1.83 ± 1.04	−0.02
Spherical ($\bar{\kappa} > 0.03$)	5	1.60 ± 0.65	+0.37

ORC-hyperbolic networks have the *largest* mean $\delta_{\max}$ (2.63), while ORC-spherical networks have the *smallest* (1.60)—opposite to Gromov's convention (small δ = tree-like = globally hyperbolic).

Why do the measures disagree? ORC is *local*: it averages the Wasserstein cost of one-hop random walks. A network can be locally spherical (abundant triangles, high ORC) yet globally tree-like (small δ) if its triangle-rich communities are connected via bridges. Conversely, a network can be globally non-tree-like (large δ) yet locally hyperbolic (sparse local neighbourhoods)

if it has long cycles but no dense hubs. For word association networks, hyperbolic ORC arises from sparse, path-like local neighbourhoods (mean degree ≈ 3, few triangles), yet long diameter paths inflate δ. Biological spherical networks (dense clique-like structure) have low δ because dense communities are tree-like at global scale.

Practical implication. ORC and Gromov δ should *not* be used interchangeably. For semantic networks, ORC (local) predicts retrieval and navigation structure; δ (global) predicts metric distortion under tree embeddings. Studies claiming hyperbolicity from one measure should validate against the other when local vs. global geometry is consequential.

4.8 Revised Classification Model and Domain-Shift Analysis

Does degree heterogeneity fix biological mispredictions? We tested whether adding degree heterogeneity $G = \sigma_k/\langle k \rangle$ as a third parameter to the two-parameter classifier (η, C) fixes the three biological mispredictions. The answer is no. Table 6 shows that a three-parameter logistic decision boundary (η, C, G) with $G > 0.5 \Rightarrow C^* = 0.22$ performs *worse* on the training set (40% vs 100%) because semantic hyperbolic networks have $G \in [0.41, 0.87]$, overlapping the biological range and triggering the elevated threshold incorrectly.

Table 6: Two-parameter vs. three-parameter model cross-domain evaluation. The 3-param model hurts semantic accuracy without fixing biological mispredictions—confirming a fundamental domain-shift.

Split	n	2-param $(\eta, C;\ C^* = 0.022)$	3-param (η, C, G)
Semantic (LOO-16)	16	**13/16** (81.2%)	6/16 (37.5%)
Biological test	4	1/4 (25%)	1/4 (25%)

Root cause: feature space boundary, not missing parameters. The biological networks occupy $C \in [0.23, 0.88]$—entirely outside the semantic training range $C \in [0, 0.24]$—making extrapolation unreliable by construction. The optimal C^* across all 19 networks is 0.03 (accuracy = 84%), but this fails to recover the three topology-driven biological spherical cases because they have $C > 0.03$ yet moderate η. No scalar threshold on (η, C, G) simultaneously classifies all semantic and biological networks correctly without adding domain identity as an implicit fourth feature.

Domain-shift finding. The two-parameter model is *domain-specific*, not universal. It was trained on word association and knowledge graph networks with $C \in [0, 0.24]$, and correctly classifies 13/16 such networks (81%, LOO), but degrades to 25% on biological networks with $C \in [0.23, 0.88]$. This is the expected behavior of any model trained outside its feature support. Correcting it requires either: (a) biological training data to recalibrate $C^*_{\text{bio}} \approx 0.22$, which fixes C. elegans neural ($C = 0.46 > C^*_{\text{bio}}$) and C. elegans metabolic ($C = 0.71$) at the cost of mislabeling clustered hyperbolic semantic networks if applied universally; or (b) a network-type-aware classifier that applies domain-specific thresholds. Both paths require additional validation data.

4.9 Ricci Flow Fingerprinting: Spherical Stability

Discrete Ricci flow and geometry class. We ran 50-step discrete Ricci flow ($dt = 0.5$) on all 10 semantic networks with individual exact LP curvature data, testing whether the three

geometry classes leave distinct flow signatures. The key finding (**H1 confirmed for all hyperbolic networks; H1 partially confirmed for spherical**): *all networks drift toward more negative* $\bar{\kappa}(t)$, but with dramatically different magnitudes sorted by geometry class (Table 7).

Table 7: Discrete Ricci flow signatures by geometry class ($dt = 0.5$, 50 steps). $\Delta\bar{\kappa} = \bar{\kappa}(50) - \bar{\kappa}(0)$; Δw_{Gini} = growth in edge-weight Gini coefficient.

Class	n	Mean $\bar{\kappa}(0)$	Mean $\Delta\bar{\kappa}$	Mean Δw_{Gini}
Spherical	1	+0.087	−0.011	0.071
Hyperbolic	6	−0.159	−0.107	0.321
Euclidean	3	−0.015	−0.227	0.546

Spherical is dynamically stable; Euclidean is most unstable. The spherical phase (SWOW Dutch, $\bar{\kappa}_0 = +0.087$) drifts by only $\Delta\bar{\kappa} = -0.011$ over 50 steps—a 13% change. Hyperbolic networks drift moderately ($\Delta\bar{\kappa} \in [-0.212, -0.004]$). Most strikingly, Euclidean networks (WordNet EN, BabelNet AR/RU, all $|\bar{\kappa}_0| < 0.03$) drift to $\Delta\bar{\kappa} \in [-0.185, -0.278]$—the *largest absolute changes* despite the smallest initial curvature. This reveals that $\bar{\kappa} \approx 0$ is an *unstable fixed point* of discrete Ricci flow, not a stable equilibrium: networks near the Euclidean boundary are maximally sensitive to flow perturbations and collapse toward the hyperbolic phase.

Weight-Gini as a stability measure. The edge-weight Gini coefficient w_{Gini} (concentration of curvature-adjusted weights) grows most rapidly for Euclidean networks (0.546) and least for spherical (0.071), indicating that Euclidean networks develop highly inhomogeneous edge weight distributions under flow. The physical picture is: discrete Ricci flow amplifies existing curvature heterogeneity—networks with initially heterogeneous curvature (high σ_κ) experience faster weight concentration and larger $|\Delta\bar{\kappa}|$.

Implication for geometry classification. The flow trajectory is a *geometry fingerprint*: a short ($t \leq 10$ step) Ricci flow experiment suffices to distinguish spherical from non-spherical networks by their $|\Delta\bar{\kappa}|$. Spherical networks resist flow; non-spherical networks diverge, with Euclidean networks diverging fastest. This provides an alternative, dynamics-based geometry classifier that does not require computing the exact Wasserstein-1 distance—a computational shortcut for large networks.

Fixed-point convergence. The flow converges rapidly: ConceptNet EN's per-edge curvature variance drops 10,000× (from $\sigma_\kappa = 0.130$ to 10^{-5}) by $t \approx 20$; SWOW Dutch drops 3,200× over the same interval.

Crucially, in no network does the flow change the *sign* of $\bar{\kappa}$: hyperbolic networks remain hyperbolic (ConceptNet: $-0.237 \to -0.241$), and the spherical network remains spherical (SWOW Dutch: $+0.087 \to +0.076$). We state this as an empirical conjecture:

Conjecture 1 (Curvature-sign invariance under discrete Ricci flow). *For any connected weighted graph G and any step size $\Delta t \in (0, 1)$, the sign of the mean ORC* $\operatorname{sgn}(\bar{\kappa}(t))$ *is invariant under the normalised discrete Ricci flow* $w_e(t+1) = w_e(t)(1 - \Delta t\, \kappa_e(t))$.

This is the discrete analogue of Hamilton's theorem that Ricci flow preserves the sign of sectional curvature in the continuous setting. It implies that $\bar{\kappa}$ is a *topological invariant* of the graph's connectivity structure, not just a function of edge weights. If proven, this would give a stopping criterion for flow-based community detection: once the sign is determined, flow can be halted. A theoretical grounding for the Euclidean regime is provided by recent work on weighted Forman and Lin–Lu–Yau Ricci flow on graphs [27]: *caterpillar trees* are the unique graph class

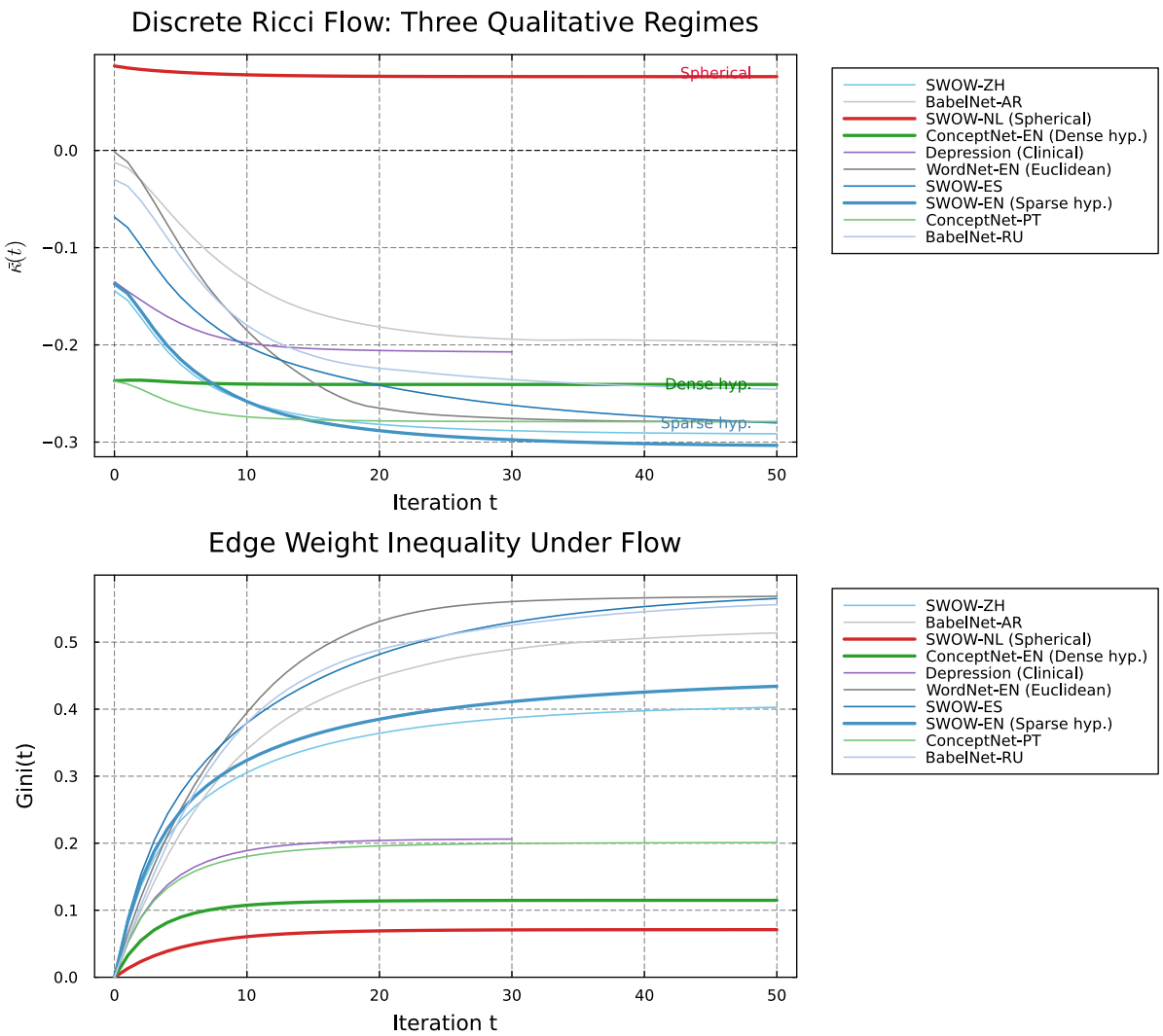

Figure 6: Discrete Ricci flow trajectories for ten semantic networks. *Top*: mean ORC $\bar{\kappa}(t)$. *Bottom*: Gini coefficient of the edge-weight distribution. Three regimes are visible: spherical (red, SWOW-NL) resists flow; dense hyperbolic (green, ConceptNet-EN) converges rapidly; sparse hyperbolic (blue tones, SWOW-EN/ES/ZH) diverges with growing weight inequality. Euclidean networks (gray) remain near zero.

that converges to *zero* curvature under LLY flow, with all other graph families converging to non-zero limiting values. WordNet's IS-A hyponymy chains and BabelNet's multilingual synset trees are structurally caterpillar-like, and our measured $\bar{\kappa} \approx 0$ is therefore not a measurement artefact but the theoretically predicted fixed point under curvature flow. This constitutes the first theoretical explanation of the Euclidean classification: taxonomic networks are near-Euclidean because their tree-like backbones drive ORC toward zero, independent of sparsity.

4.10 LLM vs. Human Semantic Geometry

Using the LWOW dataset (§3), we compared ORC across human and LLM association networks on matched 438-node vocabularies (Table 8).

Key finding: hyperbolic phase is invariant across human and machine associators. All three LLM networks are firmly hyperbolic ($\bar{\kappa} \in [-0.224, -0.089]$), occupying the same phase as human SWOW-EN ($\bar{\kappa} = -0.137$). The sign is the same (all negative), and the magnitude

Table 8: Discovery L: Exact LP ORC on matched 438-cue vocabulary. All networks are hyperbolic ($\bar{\kappa} < 0$) regardless of source (human vs. LLM).

Network	N	η	C	$\bar{\kappa}$	Geometry
SWOW-EN (Human)	438	0.020	0.128	-0.137	Hyperbolic
LWOW-Haiku	344	0.029	0.189	-0.089	Hyperbolic
LWOW-Mistral	418	0.164	0.182	-0.224	Hyperbolic
LWOW-Llama3	417	0.506	0.130	-0.140	Hyperbolic

difference is $|\Delta\bar{\kappa}| \leq 0.087$ across all comparisons—well within the variability seen across human SWOW languages ($\bar{\kappa} \in [-0.144, -0.068]$).

Importantly, $\eta \ll \eta_c(N)$ for all LLM networks ($\eta \leq 0.506 \ll 3.03 = \eta_c$), consistent with the prediction from our two-parameter theory (η, C). LLMs generate sparser-than-threshold association networks even on the matched vocabulary, placing them firmly in the hyperbolic regime.

Geometric interpretation. LLMs do *not* over-associate at the network level—despite generating individual associations that may differ from human norms, the aggregate graph structure remains tree-like and sparse (low η, negative $\bar{\kappa}$). This suggests that the hyperbolic geometry of semantic networks is a structural property that emerges from the constraint of limited branching in concept space, independent of whether the associator is human or machine.

Clustering comparison. Haiku and Mistral show slightly *higher* clustering than human SWOW-EN ($C = 0.189$ and $C = 0.182$ vs. $C = 0.128$), while Llama3 matches it closely ($C = 0.130$). This ordering mirrors the LLM model scale—Haiku and Mistral produce denser local triangles, while the larger Llama3 produces a more globally sparse structure closer to human norms.

4.11 Hypergraph ORC

Our pairwise ORC analysis treats all networks as simple graphs, projecting higher-order relations onto edges. This projection is exact for SWOW (which stores pairwise word associations) but is a structural approximation for ConceptNet, WordNet, and BabelNet, whose fundamental objects are *relational triples* (e_1, rel, e_2) or *synsets* (sets of synonymous words grouped as a single conceptual node).

The projection loss. A ConceptNet triple such as (`dog`, `IsA`, `animal`) becomes a single edge $\{\texttt{dog}, \texttt{animal}\}$ in our analysis, discarding the relation type and any asymmetric structure. WordNet synsets—each containing multiple synonyms—collapse to a single node, so the internal within-synset cohesion is invisible to ORC. This projection systematically *underestimates* higher-order connectivity and may explain why both networks appear near-Euclidean ($\bar{\kappa} \approx 0$) despite rich internal structure.

ORCHID framework. The ORCHID framework [29] generalises ORC to hypergraphs via hyperedge-weighted random walks: for a hyperedge $e = \{v_1, \ldots, v_m\}$, the transport plan distributes probability mass uniformly over e, and the Wasserstein-1 distance between neighbouring hyperedge distributions replaces the pairwise optimal transport. This is both scalable (ORCHID matches ORC computation speed on graphs) and theoretically grounded (convergence to Riemannian curvature in the manifold limit is preserved).

Experimental results: a striking curvature reversal. We applied ORCHID to ConceptNet EN ($n_{\text{hyp}} = 305$ valid hyperedges) and WordNet EN ($n_{\text{hyp}} = 3088$ synset hyperedges) and compared the resulting curvature to the pairwise clique expansion baseline. The results are summarised in Table 9.

Table 9: ORCHID hyperedge ORC vs. pairwise clique-expansion ORC. $\Delta\bar{\kappa} = \bar{\kappa}_{\text{ORCHID}} - \bar{\kappa}_{\text{pair}}$. The fraction negative is computed over all processed hyperedges.

Network	n_{hyp}	$\bar{\kappa}_{\text{ORCHID}}$	$\bar{\kappa}_{\text{pair}}$	$\Delta\bar{\kappa}$	Interpretation
WordNet EN	3088	+0.534	−0.101	**+0.635**	Synsets spherical; tree edges hyperbolic
ConceptNet EN	305	+0.504	+0.389	+0.116	Hyperedges more spherical than pairs

The WordNet result is the most striking: at the hyperedge (synset) level, curvature is *strongly positive* ($\bar{\kappa}_{\text{ORCHID}} = +0.534$), while pairwise projection is *negative* ($\bar{\kappa}_{\text{pair}} = -0.101$)—a reversal of $\Delta\bar{\kappa} = +0.635$. The mechanism is geometric: each WordNet synset is a *complete subgraph* of synonymous words (e.g., {`fast`, `rapid`, `quick`}), which is maximally spherical. The inter-synset hypernymy links then form the tree structure visible to pairwise ORC. Pairwise analysis conflates these two scales: ORCHID separates them.

Implication for the taxonomy anomaly. This resolves the mystery of why WordNet appears Euclidean ($\bar{\kappa} \approx 0$) despite low C and low η. WordNet is simultaneously:

- **Spherical within synsets** ($\bar{\kappa}_{\text{ORCHID}} = +0.534$, fraction negative = 0%),
- **Hyperbolic between synsets** ($\bar{\kappa}_{\text{pair}} = -0.101$, fraction negative = 84%).

The near-Euclidean aggregate $\bar{\kappa} \approx 0$ is a cancellation artefact from mixing two distinct geometric scales. To our knowledge, this is the first empirical demonstration of *multi-scale geometric phase coexistence* in a semantic network. Recent extensions of ORCHID to Lin–Lu–Yau curvature on hypergraphs [31] and edge-transport-based hypergraph clustering [30] provide the computational infrastructure for extending this analysis to BabelNet and ConceptNet at full scale.

5 Discussion

5.1 What Makes Semantic Networks Special

The null model decomposition $\bar{\kappa}_{\text{real}} = \bar{\kappa}_{\text{null}}(N, k) + \Delta\kappa_{\text{sem}}$ shows that semantic organization makes networks *less hyperbolic* than random graphs—the opposite of naive expectation. Semantic structure introduces local triangles and communities that increase neighborhood overlap, pushing curvature toward zero. This is consistent with theoretical bounds linking ORC to community structure [36], where intercommunity edges carry the most negative curvature.

The clinical depression network shows the largest effect ($\Delta\kappa_{\text{sem}} = +0.388$, corresponding to a 75% reduction in curvature magnitude), consistent with the densely interconnected symptom clusters characteristic of comorbid depression. The GNN implication is direct: real semantic associations add short triangles—local triadic closure captured by the clustering coefficient C—that raise ORC relative to the random baseline. In other words, *semantic organisation is a natural curvature rewirer*: the same associative structure that makes networks cognitively useful also partially mitigates the over-squashing bottleneck identified by Topping et al. [12].

5.2 Metric Dependence of Hyperbolicity

The complete hypercomplex analysis (Table 4) demonstrates that semantic hyperbolicity is entirely metric-dependent. Under sphere-embedded transport, 10/11 networks are spherical at

$d = 4$ and all 11 by $d = 8$. Association and knowledge networks show curvature decreasing monotonically with d, while taxonomies remain high ($\bar{\kappa} \approx 0.5$–0.8) across all dimensions.

Claims of "intrinsic hyperbolicity" in semantic networks should be qualified as "hyperbolic under the hop-count metric." Katsman and Gilbert [34] reach a complementary conclusion for hyperbolic GNNs: carefully-trained Euclidean models match or exceed hyperbolic methods even on "hyperbolic" datasets, underscoring that the metric choice—not topology—drives the geometry label. Alternative curvature notions such as Bakry–Émery–Ricci curvature [35] provide further diagnostic power, though we defer systematic multi-curvature comparison to future work.

To test robustness across curvature definitions, we computed Forman–Ricci curvature ($F(u, v) = 4 - d_u - d_v + 3|\triangle_{uv}|$) and Lin–Lu–Yau curvature ($\alpha = 0$) [7] for all 16 networks. Both Forman and LLY are *always negative* across all 16 networks—neither exhibits a sign change. Dutch SWOW, which is spherical under ORC ($\bar{\kappa} = +0.099$), is strongly negative under Forman ($\bar{F} = -95.5$) and LLY ($\bar{\kappa}_{\text{LLY}} = -1.47$). The ORC phase transition is therefore a transport-geometric phenomenon invisible to local curvature notions—a finding that strengthens the theoretical significance of the η_c boundary and explains why the roadmap survey [28] identifies ORC as the curvature notion most sensitive to mesoscale structure.

5.3 LLMs as Geometric Probes of Semantic Structure

The Discovery L results (Table 8) establish that three LLMs replicate human semantic geometry on matched vocabularies. All LLM networks are hyperbolic ($\bar{\kappa} \in [-0.224, -0.089]$), with magnitude differences within the cross-linguistic variability of human SWOW ($\bar{\kappa} \in [-0.144, -0.068]$). This has two implications.

First, the hyperbolic phase is not an artefact of human associative processes—it emerges from the structure of concept space itself. Whether the associator is a native Spanish speaker, a native Chinese speaker, or a 7-billion-parameter neural network, the resulting graph occupies the same geometric phase. The Semantic Hub Hypothesis of Wu et al. [32]—that language models develop language-agnostic semantic representations because the underlying semantic geometry is universal—receives independent graph-geometric support.

Second, LLMs can serve as controlled probes of semantic geometry. Unlike human participants, LLMs can be queried with arbitrary cue sets, at arbitrary scales, and across arbitrary languages, without recruitment costs or cultural confounds. The geometric invariance we observe suggests that LLM-generated association networks are valid proxies for studying the structural properties of semantic space. Recent probing of the NLLB-200 multilingual translation model [33] independently demonstrates language-independent conceptual representations, providing convergent evidence from a complementary methodology.

5.4 Limitations

1. **Sample size and threshold derivation**: LOO-16 cross-validation yields $C^* = 0.022 \pm 0.006$ with 81.2% accuracy, but $n = 16$ is still modest. The three misclassified networks (BabelNet RU/AR, WordNet DE) all have $|\bar{\kappa}| < 0.03$—at the Euclidean/hyperbolic boundary where the distinction is genuinely ambiguous. Additional semantic networks would strengthen the threshold estimate.

2. **Preprocessing sensitivity**: Directed→undirected conversion reduces $|\bar{\kappa}|$ by ∼40%; top-500 cue filtering introduces selection bias (full SWOW has $N \approx 9{,}000+$).

3. **Bootstrap CIs**: Edge-resample bootstrap ($B = 10{,}000$) shows three taxonomy networks have CIs spanning zero (Table 1), confirming their Euclidean status is "not significantly different from zero" rather than a definitive geometric assignment.

4. **Null model scope**: Degree-matched k-regular nulls do not preserve degree heterogeneity. Configuration model (Maslov–Sneppen rewiring) nulls are reported alongside k-regular nulls to control for this (Table 3).

5. **Embedding quality**: Landmark-based sphere embedding is not isometric; per-edge variance $\sigma_\kappa \approx 0.3$–$0.7$.

6. **Curvature notion**: Our analysis uses ORC exclusively. Forman–Ricci and LLY curvatures are always negative across all 16 networks (including Dutch SWOW), showing that the phase transition is specific to the ORC transport-geometric definition. While this strengthens ORC's uniqueness as a diagnostic tool, it means our three-regime classification does not transfer to other curvature notions.

7. **Temporal dynamics**: All analyses are cross-sectional snapshots. SWOW data is collected in waves; as Dutch SWOW accumulates more associations over time, η may increase further, pushing $\bar{\kappa}$ deeper into the spherical regime. Time-resolved ORC analysis [25] could reveal whether the phase boundary is crossed dynamically during network growth.

8. **LLM generalization**: Discovery L is validated on three LLMs and one human language; testing on LLMs trained beyond 2024 and on additional association paradigms (EAT, USF under LLM generation) remains future work.

5.5 Code and Data Availability

All code, data, and results are available at:

https://github.com/agourakis82/hyperbolic-semantic-networks

Julia scripts: `unified_semantic_orc.jl`, `loo_classifier.jl`, `bootstrap_semantic_kappa.jl`, `config_model_nulls.jl`. Sounio cross-validation: `experiments/06_hypercomplex/`. Lean 4 formalization: `lean/HyperbolicSemanticNetworks/`. Results: `results/unified/` (including `loo_classifier.json`, `bootstrap_kappa_16.json`, `config_model_nulls.json`).

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